Cooperative control of a tethered drone and surface vessel for sustained presence at sea

Tomás Grossinho Esteves

Thesis to obtain the Master of Science Degree in

Electrical and Computer Engineering

Supervisor(s): Prof. David Alexandre Cabecinhas
Prof. Rita Maria Mendes de Almeida Correia da Cunha

Examination Committee

Prof. Meysam Basiri

January 2026

Declaration

I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

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Chapter 1

Introduction

This chapter introduces the thesis, presenting the context, motivation, and objectives of the research, followed by an overview of the system architecture and the expected scientific and engineering contributions.

1.1 Motivation

1.2 Problem Statement

1.3 System Overview

1.4 Expected Contributions

1.5 Thesis Structure

Chapter 2

State of the Art and Related Work

This chapter reviews the literature on modeling, estimation, control, and coordination of unmanned systems, focusing on cooperative surface-aerial systems, tethered platforms, and active tension control.

2.1 Modeling and System Identification

2.1.1 White-Box, Grey-Box, and Black-Box Modeling

2.1.2 Tether and Cable Dynamics Modeling

2.1.3 Optimization-Based Estimation

2.1.4 Related Work on System Identification

2.2 Control Strategies for Autonomous Vehicles

2.2.1 Classical and Modern Control

2.2.2 Predictive and Optimization-Based Control

2.2.3 Active Winch and Tension Control

2.2.4 Related Work on Tethered Vehicle Control

2.3 Cooperative Coordination and Systems

2.3.1 Centralized vs. Distributed Coordination

2.3.2 UAV-USV Cooperative Systems

2.3.3 Related Work on Maritime Cooperative Robotics

2.4 Path and Trajectory Planning

2.4.1 Offline and Online Planning

2.4.2 Related Work on Trajectory Planning

Chapter 3

System Modeling and Validation

This chapter details the coordinate systems, kinematics, and dynamic models of the vehicles and their physical link. It covers the WAM-V USV, the x500 UAV, and the tether-winch coupling. For each component, the high-fidelity simulation model representing the physical plant is presented, followed by the simplified control-oriented mathematical model and its validation against simulator telemetry.

3.1 Coordinate Systems and Kinematics

To describe the motion and coupling of the cooperative marine-aerial system, it is necessary to establish the reference coordinate frames and the transformations between them. In this work, three primary coordinate systems are defined, as illustrated in Fig. 3.1:

- **Inertial Reference Frame** $\{\mathcal{I}\}$: A local flat-Earth coordinate system defined by the origin $\mathcal{O}_I$ and axes $\{\mathbf{x}_I, \mathbf{y}_I, \mathbf{z}_I\}$. This frame is oriented according to the East-North-Up (ENU) convention, where $\mathbf{x}_I$ points East, $\mathbf{y}_I$ points North, and $\mathbf{z}_I$ points vertically upward, parallel to the gravity vector.

- **USV Body-Fixed Reference Frame** $\{\mathcal{B}_v\}$: Centered at the WAM-V catamaran's center of gravity $\mathcal{O}_v$ with axes $\{\mathbf{x}_v, \mathbf{y}_v, \mathbf{z}_v\}$. The longitudinal axis $\mathbf{x}_v$ points forward (surge direction), the lateral axis $\mathbf{y}_v$ points to port/left (sway direction), and the vertical axis $\mathbf{z}_v$ points upward (heave direction).

- **UAV Body-Fixed Reference Frame** $\{\mathcal{B}_a\}$: Centered at the x500 quadrotor's center of gravity $\mathcal{O}_a$ with axes $\{\mathbf{x}_a, \mathbf{y}_a, \mathbf{z}_a\}$. The axis $\mathbf{x}_a$ points forward, $\mathbf{y}_a$ points to the left, and $\mathbf{z}_a$ points upward.

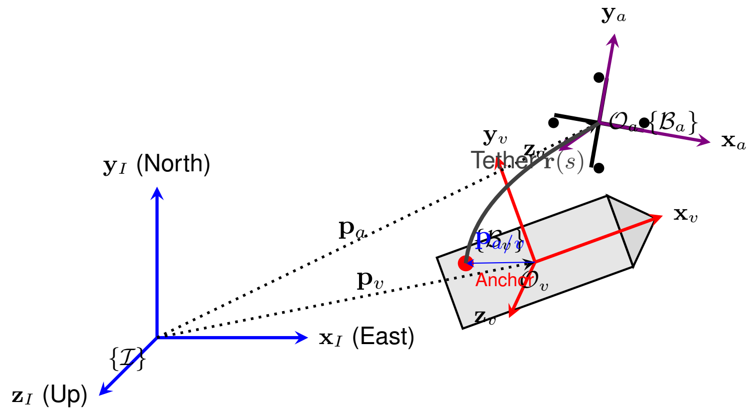

Figure 3.1: Schematic representation of the reference coordinate frames $\{\mathcal{I}\}$, $\{\mathcal{B}_v\}$, $\{\mathcal{B}_a\}$, and the physical anchor connection.

The position vectors of the USV center of gravity and the UAV center of gravity in the inertial frame $\{\mathcal{I}\}$ are defined as $\mathbf{p}_v = [x_v, y_v, z_v]^T$ and $\mathbf{p}_a = [x_a, y_a, z_a]^T$, respectively. Under nominal operations, the USV moves strictly on the water surface, meaning that $z_v = 0$ is assumed.

The physical tether anchor on the WAM-V deck is located at a fixed offset vector $\mathbf{p}_{a/v} = [x_{a/v}, y_{a/v}, z_{a/v}]^T$ relative to the USV body-fixed frame $\{\mathcal{B}_v\}$. The global position of this anchor point in the inertial frame is given by:

$$\mathbf{p}_{anchor} = \mathbf{p}_v + \mathbf{R}_v(\psi)\mathbf{p}_{a/v} \tag{3.1}$$

where $\mathbf{R}_v(\psi) \in SO(3)$ is the rotation matrix representing the orientation of the USV.

Since the USV roll and pitch motions are generally small and neglected in the control-oriented model, the rotation from $\{\mathcal{B}_v\}$ to $\{\mathcal{I}\}$ is parameterized by the yaw heading angle ψ, representing a planar rotation about the vertical axis $\mathbf{z}_I$:

$$\mathbf{R}_v(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{3.2}$$

For the UAV, the orientation of $\{\mathcal{B}_a\}$ relative to $\{\mathcal{I}\}$ is parameterized by the standard Euler angles: roll ϕ, pitch θ, and yaw ψ_a, using the Z-Y-X rotation sequence (yaw, then pitch, then roll). The resulting rotation matrix $\mathbf{R}_a(\phi, \theta, \psi_a) \in SO(3)$ is:

$$\mathbf{R}_a(\phi, \theta, \psi_a) = \begin{bmatrix} \cos\psi_a \cos\theta & \cos\psi_a \sin\theta \sin\phi - \sin\psi_a \cos\phi & \cos\psi_a \sin\theta \cos\phi + \sin\psi_a \sin\phi \\ \sin\psi_a \cos\theta & \sin\psi_a \sin\theta \sin\phi + \cos\psi_a \cos\phi & \sin\psi_a \sin\theta \cos\phi - \cos\psi_a \sin\phi \\ -\sin\theta & \cos\theta \sin\phi & \cos\theta \cos\phi \end{bmatrix} \tag{3.3}$$

This mathematical kinematic relationship allows forces and positions computed in the body-fixed frames of each vehicle to be mapped to the inertial space, which is required to enforce tether constraints and ensure coordinated trajectory tracking.

3.2 WAM-V Surface Vessel Modeling and Validation

3.2.1 USV Kinematics

3.2.2 High-Fidelity Simulation Model (Gazebo Hydrodynamics)

3.2.3 Control-Oriented Dynamic Model (Fossen 3-DOF)

3.2.4 Thruster Allocation Mapping

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3.3 x500 Quadrotor Modeling and Validation

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3.3.3 Control-Oriented Model (Attitude Lag)

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3.4.2 Control-Oriented Geometric Constraints

3.4.3 Virtual Winch Kinematics and Spooling Modes

3.4.4 Tether Force and Slack Validation

Chapter 4

Cooperative Control Design

This chapter describes the controller design for the coupled system. It details the principles of Nonlinear Model Predictive Control (NMPC), the specific optimization formulations for the UAV and the USV tracking control, and the active winch spooling control strategy.

4.1 Nonlinear Model Predictive Control Principles

4.2 UAV NMPC Optimization Problem

4.3 USV NMPC Optimization Problem

4.4 Active Winch Control Strategy

Chapter 5

Simulation and Software Environment

This chapter describes the software stack, nodes, and co-simulation setups used to implement and test the cooperative tethered control system. It details the code generation pipeline, the C++ ROS 2 nodes, and the integration of the MoorDyn physics engine and virtual winch controller within the Gazebo simulation environment.

5.1 Software Environment and Code Generation

5.2 ROS 2 Software Architecture

5.3 Co-Simulation Setup in Gazebo Harmonic

5.4 MoorDyn and Winch Integration

Chapter 6

Results and Discussion

This chapter presents the closed-loop simulation results and evaluations. It assesses the trajectory tracking performance of the individual controllers, demonstrates the cooperative tracking capabilities of the coupled UAV-USV system, validates the active winch control modes under wave perturbations, and tests the robustness of the safety constraints under external disturbances.

6.1 Simulation Setup and Environmental Scenarios

6.2 USV Closed-Loop Tracking Performance

6.2.1 Path-Following and Heading Control

6.2.2 Thruster Mapper Output

6.3 Cooperative UAV-USV Target Tracking (Dynamic Anchor)

6.4 Active Winch Control Performance

6.4.1 Geometric Spooling Mode

6.4.2 Tension-Feedback PD Mode

6.5 Constraint Enforcement and Robustness Analysis

6.5.1 Enforcement of Max Tether Length Constraint

6.5.2 Robustness to Wind and Wave Disturbances

Chapter 7

Conclusions

This chapter concludes the dissertation, summarizing the key achievements and proposing directions for future research.

7.1 Conclusions

7.2 Future Work

Bibliography